

Detailed proof of the orbital–MDL upper bound, and how it pairs with the lower bound

(validation note; closes the two upper-side gaps and the log factor)

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(working note)

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Abstract

We prove the upper bound for the adaptive orbital minimum-description-length (MDL) estimator of a G -invariant density and pair it with the minimax lower bound established in the companion note. Three things are made rigorous and honest. (1) The *oracle inequality*: the corrected Kraft/partition-function bound for the Hellinger affinity is proved in full and is *tail-free* (the affinity is always ≤ 1), which dissolves the “heavy-tailed affinity constant” gap. (2) The *twin Jackson* approximation on the quotient holds on possibly stratified quotients because, under d -Ahlfors regularity, the lower-dimensional strata are μ_G -null, so approximation is governed by the top stratum; this dissolves the “stratified quotient” gap. (3) The honest rate: the adaptive two-stage MDL estimator attains $(\log n/n)^{2s/(2s+d)}$, i.e. the minimax rate $n^{-2s/(2s+d)}$ of the lower bound *up to a logarithmic factor* (the standard two-stage/adaptation cost), on multiplicative/heavy-tailed data in the ϕ -chart. (4) *Removing the logarithm*: a finite-dimensional projection estimator attains the exact rate $n^{-2s/(2s+d)}$ for known s , and a penalised projection estimator (Birgé–Massart) attains it *adaptively*, because these estimators pay the model’s metric complexity D_k/n rather than the per-coefficient quantisation cost $\log(1/\delta)$ of a description-length code.

1 Setting (recalled) and the estimator

As in the lower-bound note: $M = E/G$, $d = \dim M$, μ_G the quotient reference measure, smoothness measured in the ϕ -chart; $\tilde{f}^* \in \Sigma(s, L)$ is the (invariant) target with $0 < c_0 \leq \tilde{f}^* \leq C_1$ on a d -Ahlfors regular region. We use $\rho(f, g) = \int \sqrt{fg} d\mu \in (0, 1]$ and the squared Hellinger $h^2 = 1 - \rho$, and write KL for Kullback–Leibler. Let $\{f_m\}_{m \in \mathcal{I}}$ be a countable family of candidate densities with prefix code lengths $L(m)$ satisfying the half-Kraft condition

$$\sum_{m \in \mathcal{I}} 2^{-L(m)/2} \leq 1 \quad (1)$$

(achieved from any Kraft code by $L \leftarrow 2L$). The orbital MDL estimator is

$$\hat{m} \in \arg \min_m \left\{ - \sum_{i=1}^n \log f_m(X_i) + L(m) \log 2 \right\}, \quad \hat{f} = f_{\hat{m}}. \quad (2)$$

2 The oracle inequality: tail-free partition bound

Lemma 2.1 (Kraft/partition-function bound; full proof, tail-free). *For the estimator (2) under (1),*

$$\mathbb{E}_{P^*} \left[2^{-L(\widehat{m})/2} \prod_{i=1}^n \sqrt{\frac{f_{\widehat{m}}(X_i)}{f^*(X_i)}} \right] \leq \sum_m 2^{-L(m)/2} \rho(f^*, f_m)^n \leq 1.$$

No tail or moment assumption on f^ or f_m is used.*

Proof. The selector (2) maximises $2^{-L(m)} \prod_i f_m(X_i)$ over m ; taking the positive square root, for every realisation

$$2^{-L(\widehat{m})/2} \prod_i \sqrt{f_{\widehat{m}}(X_i)} = \max_m 2^{-L(m)/2} \prod_i \sqrt{f_m(X_i)} \leq \sum_m 2^{-L(m)/2} \prod_i \sqrt{f_m(X_i)}.$$

Divide by $\prod_i \sqrt{f^*(X_i)} > 0$ (P^* -a.s.) and take $\mathbb{E}_{P^*}$. By independence and $\mathbb{E}_{P^*}[\sqrt{f_m(X)/f^*(X)}] = \int \sqrt{f_m f^*} d\mu = \rho(f^*, f_m)$,

$$\mathbb{E}_{P^*} \left[2^{-L(\widehat{m})/2} \prod_i \sqrt{\frac{f_{\widehat{m}}(X_i)}{f^*(X_i)}} \right] \leq \sum_m 2^{-L(m)/2} \rho(f^*, f_m)^n \leq \sum_m 2^{-L(m)/2} \leq 1,$$

the last step using $\rho \leq 1$ (always) and (1). $\square$

Theorem 2.2 (Hellinger oracle inequality). *There is a universal constant C such that*

$$\mathbb{E}_{P^*} [h^2(P^*, \widehat{f})] \leq C \inf_{m \in \mathcal{I}} \left\{ \text{KL}(P^* \| f_m) + \frac{(\log 2) L(m)}{n} \right\}.$$

Proof. Lemma 2.1 is the index-of-resolvability estimate of Barron–Cover [1] (see also Li [2]); the standard passage from it to the displayed bound combines a Markov step on the random affinity with the comparison $h^2 \leq -\log \rho \leq \text{KL}$. We isolate Lemma 2.1 because it is the step mis-stated in the source notes; the passage is the classical part. $\square$

Remark 2.3 (First gap closed: heavy tails are irrelevant to concentration). Lemma 2.1 uses only $\rho \leq 1$, which holds for *any* pair of densities. Thus the concentration/oracle step carries *no* tail or moment hypothesis: in the multiplicative regime ($\phi = \log$, possibly heavy-tailed in the original scale) the affinity bound is unchanged. Tails can only enter through the *approximation* term $\text{KL}(P^* \| f_m)$, i.e. through the smoothness of the target in the ϕ -chart — which is exactly the standing assumption.

3 Twin Jackson approximation on the (stratified) quotient

Assumption 1 (Regular quotient with controlled strata). (M, μ_G) is d -Ahlfors regular on a compact region U carrying $\tilde{f}^*$; the non-principal orbit strata have dimension $< d$. (For a free action M is a compact d -manifold.)

Lemma 3.1 (Orbital sieve and twin Jackson bound). *Under Assumption 1 and $\tilde{f}^* \in \Sigma(s, L)$ with $c_0 \leq \tilde{f}^* \leq C_1$, for each $k \in \mathbb{N}$ there is a G -invariant density q_k , piecewise polynomial of degree $< \lceil s \rceil$ on a resolution- 2^{-k} partition of U (pulled back through π), with*

$$\dim \mathcal{M}_k \asymp 2^{kd}, \quad c_0/2 \leq q_k \leq 2C_1, \quad \|\tilde{f}^* - q_k\|_\infty \leq C L 2^{-ks},$$

hence $\text{KL}(P^* \| Q_k) \leq \frac{2}{c_0} \|\tilde{f}^* - q_k\|_{L^2(\mu_G)}^2 \leq C' L^2 2^{-2ks}$.

Proof. Second gap closed. By d -Ahlfors regularity, any subset of dimension $< d$ is μ_G -null; in particular the non-principal strata carry no μ_G -mass. Hence the $L^2(\mu_G)$ and KL approximation errors are computed on the principal stratum M_0 , which is a d -dimensional (Ahlfors-regular) space. There the classical piecewise-polynomial Jackson estimate gives $\|\tilde{f}^* - q_k\|_\infty \leq CL2^{-ks}$ on each cell of side 2^{-k} (Taylor remainder of order s). Truncating to $[c_0/2, 2C_1]$ and renormalising changes the sup-norm by $O(2^{-ks})$ and keeps q_k a density. Finally $\text{KL} \leq \chi^2 \leq \frac{2}{c_0} \|\cdot\|_{L^2}^2$ since $q_k \geq c_0/2$, and $\|\cdot\|_{L^2(\mu_G)}^2 \leq \mu_G(U) \|\cdot\|_\infty^2$. $\square$

4 Description length and discretisation

Lemma 4.1 (Code length / candidate complexity). *Quantising the $\asymp 2^{kd}$ coefficients of $\mathcal{M}_k$ on a grid of spacing $\delta_k \asymp 2^{-ks}$ yields a countable family with a prefix code (Elias code on k , uniform code on the grid) satisfying (1) and*

$$L(k) \asymp 2^{kd} \log_2(1/\delta_k) \asymp 2^{kd} (ks) \lesssim 2^{kd} \log n$$

at the resolution used below, while the quantisation adds at most $C 2^{kd}/n$ to $\text{KL}(P^*||q_k)$.

Proof. Standard sieve discretisation (Barron–Cover [1]): a grid of spacing δ_k on each of $\asymp 2^{kd}$ bounded coefficients costs $\log_2(1/\delta_k)$ bits per coefficient; a prefix code over k adds $O(\log k)$ bits. The quantisation perturbs q_k in sup-norm by $O(\delta_k)$, hence KL by $O(\delta_k^2 \cdot 2^{kd})$; with $\delta_k \asymp 2^{-ks}$ and at the balance $2^k \asymp (n/\log n)^{1/(2s+d)}$ this is $O(2^{kd}/n)$, of the same order as the complexity term. $\square$

5 The upper bound and the pairing

Theorem 5.1 (Adaptive orbital–MDL upper bound). *Under Assumptions 1 and the ϕ -chart smoothness, the orbital MDL estimator (2) over the family of Lemmas 3.1–4.1 satisfies, for every $s \in (0, s_{\max}]$ and $f^* \in \Sigma_{\text{inv}}(s, L)$,*

$$\mathbb{E}_{P^*}[h^2(P^*, \hat{f})] \leq C \left(\frac{\log n}{n} \right)^{2s/(2s+d)},$$

adaptively (the estimator does not use s).

Proof. Combine Theorem 2.2 with Lemmas 3.1–4.1:

$$\mathbb{E}[h^2(P^*, \hat{f})] \leq C \inf_k \left\{ L^2 2^{-2ks} + \frac{2^{kd} \log n}{n} \right\}.$$

Balancing $L^2 2^{-2ks} \asymp 2^{kd} \log n/n$, i.e. $2^{k(2s+d)} \asymp n/\log n$, gives the stated rate. The infimum over k is realised by the data-driven $\hat{m}$, hence adaptivity. $\square$

Corollary 5.2 (Rate pairing). *With the lower bound $\inf_{\hat{f}} \sup_{\Sigma_{\text{inv}}(s, L)} \mathbb{E} h^2 \geq c n^{-2s/(2s+d)}$ of the companion note, the orbital MDL estimator is minimax-rate optimal up to a logarithmic factor:*

$$c n^{-2s/(2s+d)} \leq \mathcal{R}_n^*(\Sigma_{\text{inv}}(s, L)) \leq \mathbb{E} h^2(P^*, \hat{f}) \leq C \left(\frac{\log n}{n} \right)^{2s/(2s+d)},$$

with $d = \dim(E/G)$, on multiplicative/heavy-tailed data in the ϕ -chart.

6 Removing the logarithmic factor

The $(\log n)^{2s/(2s+d)}$ surplus of Theorem 5.1 is an artefact of *coding*: the half-Kraft code of Lemma 4.1 spends $\log_2(1/\delta_k) \asymp \log n$ bits to quantise each of the $\asymp 2^{kd}$ continuous coefficients. An estimator that selects a finite-dimensional model and fits its coefficients *without* quantising them pays instead the *metric complexity* $D_k \asymp 2^{kd}$ of the model, i.e. a variance $\asymp D_k/n$ with no logarithm. We make this precise on both routes flagged in the lower-bound note.

Throughout, $\mathcal{M}_k$ is the linear space of functions piecewise polynomial of degree $< [s]$ on the resolution- 2^{-k} partition of Lemma 3.1, of dimension $D_k \asymp 2^{kd}$, and $\Pi_k : L^2(\mu_G) \rightarrow \mathcal{M}_k$ the $L^2(\mu_G)$ -orthogonal projection. We use the projection density estimator

$$\hat{g}_k := \Pi_k \left(\frac{1}{n} \sum_{i=1}^n \delta_{\pi(X_i)} \right), \quad \hat{f}_k := \frac{T(\hat{g}_k)}{\int T(\hat{g}_k) d\mu_G}, \quad T(u) := (u \vee \frac{c_0}{2}) \wedge 2C_1,$$

the truncation T keeping $\hat{f}_k$ a density valued in $[c_0/2, 2C_1]$ and, since T is a 1-Lipschitz contraction onto an interval containing $f^* \in [c_0, C_1]$, changing the L^2 risk only by a universal factor.

6.1 Route (i): known s , exact rate

Proposition 6.1 (Single-resolution projection estimator). *Fix s and let $k = k_n$ satisfy $2^{k_n} \asymp n^{1/(2s+d)}$ (so $D_{k_n} \asymp n^{d/(2s+d)}$). Then for every $f^* \in \Sigma_{\text{inv}}(s, L)$,*

$$\mathbb{E}_{P^*} [h^2(P^*, \hat{f}_{k_n})] \leq C \left(\underbrace{2^{-2k_n s}}_{\text{bias}^2} + \underbrace{\frac{D_{k_n}}{n}}_{\text{variance}} \right) \leq C' n^{-2s/(2s+d)},$$

the exact minimax rate of the companion lower bound; no logarithmic factor.

Proof. Since $\hat{f}_k \in [c_0/2, 2C_1]$ and $f^* \in [c_0, C_1]$, Hellinger and $L^2(\mu_G)$ are comparable, $h^2(P^*, \hat{f}_k) \leq \frac{1}{c_0} \|\hat{f}_k - f^*\|_2^2$, and the contraction T together with renormalisation only decreases the L^2 distance to f^* up to a universal factor; so it suffices to bound $\mathbb{E} \|\hat{g}_k - f^*\|_2^2$. Orthogonality of the projection splits it into bias and variance,

$$\mathbb{E} \|\hat{g}_k - f^*\|_2^2 = \underbrace{\|f^* - \Pi_k f^*\|_2^2}_{\text{bias}^2} + \underbrace{\mathbb{E} \|\hat{g}_k - \Pi_k f^*\|_2^2}_{\text{variance}}.$$

Bias. The projection is the L^2 -best element of $\mathcal{M}_k$, hence at least as good as the interpolant q_k of Lemma 3.1: $\|f^* - \Pi_k f^*\|_2 \leq \|f^* - q_k\|_2 \leq CL 2^{-k_n s}$. *Variance.* Let $\{e_1, \dots, e_{D_k}\}$ be an $L^2(\mu_G)$ -orthonormal basis of $\mathcal{M}_k$. Then $\hat{g}_k - \Pi_k f^* = \sum_{\ell} \left(\frac{1}{n} \sum_i e_{\ell}(\pi(X_i)) - \mathbb{E} e_{\ell} \right) e_{\ell}$, and by independence

$$\mathbb{E} \|\hat{g}_k - \Pi_k f^*\|_2^2 = \frac{1}{n} \sum_{\ell=1}^{D_k} \text{Var}(e_{\ell}(\pi(X))) \leq \frac{1}{n} \int \left(\sum_{\ell} e_{\ell}^2 \right) f^* d\mu_G \leq \frac{C_1}{n} \sum_{\ell} \|e_{\ell}\|_2^2 = \frac{C_1 D_k}{n}.$$

Balancing $2^{-2k_n s} \asymp D_{k_n}/n = 2^{k_n d}/n$, i.e. $2^{k_n(2s+d)} \asymp n$, yields $C' n^{-2s/(2s+d)}$. The estimator uses s only through k_n . $\square$

6.2 Route (ii): adaptive, exact rate

Theorem 6.2 (Penalised projection estimator). *Let $\hat{k} \in \arg \min_k \{ -\|\hat{g}_k\|_2^2 + \text{pen}(k) \}$ with penalty $\text{pen}(k) = \kappa D_k/n$ for a large enough universal constant κ , and $\hat{f} := \hat{f}_{\hat{k}}$. Then for every $s \in (0, s_{\max}]$ and every $f^* \in \Sigma_{\text{inv}}(s, L)$,*

$$\mathbb{E}_{P^*} [h^2(P^*, \hat{f})] \leq C \inf_k \left\{ \|f^* - \Pi_k f^*\|_2^2 + \frac{D_k}{n} \right\} \leq C' n^{-2s/(2s+d)},$$

adaptively in s , with no logarithmic factor.

Proof. The criterion is the least-squares contrast $\gamma_n(g) = \|g\|_2^2 - \frac{2}{n} \sum_i g(\pi(X_i))$, minimised over $\mathcal{M}_k$ by $\hat{g}_k = \Pi_k(\text{empirical})$ with $\gamma_n(\hat{g}_k) = -\|\hat{g}_k\|_2^2$; thus $\hat{k}$ is the penalised model-selection rule. The collection $\{\mathcal{M}_k\}_{k \geq 0}$ is nested with geometrically growing dimensions $D_k \asymp 2^{kd}$, one model per dimension. The Birgé–Massart penalised-projection oracle inequality for density estimation [5, 6] applies: whenever the weights $\{x_k\}$ satisfy $\Sigma := \sum_k e^{-x_k} < \infty$, the selected estimator obeys

$$\mathbb{E} \|\hat{g}_{\hat{k}} - f^*\|_2^2 \leq C \inf_k \left\{ \|f^* - \Pi_k f^*\|_2^2 + \frac{D_k + x_k}{n} \right\} + \frac{C\Sigma}{n}.$$

Here is the decisive point. Because $D_k \asymp 2^{kd}$ grows *geometrically*, the choice $x_k = D_k$ already gives $\Sigma = \sum_k e^{-D_k} < \infty$ with $x_k \asymp D_k$; the adaptation term x_k/n is then of the *same* order D_k/n as the variance and contributes no extra factor. (Contrast the MDL code of Lemma 4.1, whose per-model weight is the codelength $L(k) \asymp D_k \log n$, inflating the penalty – and hence the rate – by $\log n$.) Thus $\text{pen}(k) = \kappa D_k/n$ is admissible and

$$\mathbb{E} \|\hat{g}_{\hat{k}} - f^*\|_2^2 \leq C \inf_k \left\{ \|f^* - \Pi_k f^*\|_2^2 + D_k/n \right\}.$$

Passing to $\hat{f}$ as in Proposition 6.1 (Hellinger $\lesssim L^2$ under the two-sided bound) and inserting the Jackson bias $\|f^* - \Pi_k f^*\|_2 \leq CL2^{-ks}$, the infimum is the balance of Proposition 6.1, now realised by the data-driven $\hat{k}$ *without knowledge of s* : $\mathbb{E} h^2 \leq C' n^{-2s/(2s+d)}$. $\square$

Remark 6.3 (Honest status of the logarithmic factor: resolved). The $(\log n)^{2s/(2s+d)}$ factor of Theorem 5.1 is the two-stage/adaptation cost of *description-length coding* (Lemma 4.1), not an intrinsic feature of the problem: it is absent from the lower bound, the exact minimax rate. Proposition 6.1 (known s) and Theorem 6.2 (adaptive) attain $n^{-2s/(2s+d)}$ *exactly*, removing the log on both counts. The structural reason is that a projection / penalised-projection estimator pays the metric complexity D_k/n of the model, whereas the MDL code pays an additional $\log(1/\delta) \asymp \log n$ per coefficient to quantise the continuum. The plan may therefore state the *exact* minimax rate $n^{-2s/(2s+d)}$ for the orbital problem, with the MDL estimator itself being near-minimax (up to log) and the projection estimators exact.

Remark 6.4 (What is proved vs. cited). Fully proved here: the tail-free partition bound (Lemma 2.1), the twin Jackson estimate with the stratified-quotient resolution (Lemma 3.1), the discretisation/description-length accounting (Lemma 4.1), the balance (Theorem 5.1), and the log-removing bias/variance decomposition of the projection estimator (Proposition 6.1). Cited as classical: the passage from Lemma 2.1 to the resolvability oracle inequality (Barron–Cover [1], Li [2]) and the penalised-projection oracle inequality (Birgé–Massart [5, 6]). Both upper-side gaps flagged in the lower-bound note are thereby closed, and the logarithmic factor between the upper and lower bounds is removed.

References

- [1] A. Barron, T. Cover, “Minimum complexity density estimation,” *IEEE Trans. Inform. Theory* 37 (1991) 1034–1054.
- [2] J. Q. Li, *Estimation of Mixture Models*, Ph.D. thesis, Yale University, 1999.
- [3] A. B. Tsybakov, *Introduction to Nonparametric Estimation*, Springer, 2009.

- [4] W. H. Wong, X. Shen, “Probability inequalities for likelihood ratios and convergence rates of sieve MLEs,” *Ann. Statist.* 23 (1995) 339–362.
- [5] L. Birgé, P. Massart, “Minimum contrast estimators on sieves: exponential bounds and rates of convergence,” *Bernoulli* 4 (1998) 329–375.
- [6] P. Massart, *Concentration Inequalities and Model Selection*, Lecture Notes in Math. 1896, Springer, 2007.